

EF Core 8.0 Guided Hands-On Exercises — Answer Key

Lab 5: Retrieving Data from the Database

Scenario	The store wants to display product details on the dashboard.
Objective	Use Find, FirstOrDefault, and ToListAsync to retrieve data.

Step 1: Retrieve All Products

```
var products = await context.Products.ToListAsync();
foreach (var p in products)
    Console.WriteLine($"{p.Name} - ■{p.Price}");
```

Explanation: ToListAsync() executes a `SELECT * FROM Products` against the database and materializes every row into a `List<Product>` in memory. Use this when you need the full result set.

Expected output (based on Lab 4's seeded data):

```
Laptop - ■75000
Rice Bag - ■1200
```

Step 2: Find by ID

```
var product = await context.Products.FindAsync(1);
Console.WriteLine($"Found: {product?.Name}");
```

Explanation: FindAsync looks up an entity by its primary key. It first checks the **change tracker / in-memory cache** — if the entity with that key is already tracked (e.g. just inserted in this same context instance), no database query is made at all. Otherwise it queries the database once. This makes FindAsync the most efficient option for primary-key lookups.

Step 3: FirstOrDefault with Condition

```
var expensive = await context.Products
    .FirstOrDefaultAsync(p => p.Price > 50000);
Console.WriteLine($"Expensive: {expensive?.Name}");
```

Explanation: FirstOrDefaultAsync translates the lambda into a SQL WHERE clause and returns the *first matching row*, or null if none match — it never throws, unlike FindAsync(). Given the seeded data (Laptop = 75000), this returns the Laptop product.

Comparison of Retrieval Methods

Method	Use case	If no match
ToListAsync()	Get all rows / a filtered list	Returns empty list
FindAsync(key)	Lookup by primary key (checks tracker first)	Returns null

FirstOrDefaultAsync(predicate)	First row matching a condition	Returns null
FirstAsync(predicate)	First row matching a condition, must exist	Throws exception
SingleOrDefaultAsync(predicate)	Exactly one expected match	Returns null (throws if >1)

Key Takeaway

EF Core offers several ways to read data — choose `ToListAsync` for full/filtered lists, `FindAsync` for fast primary-key lookups, and `FirstOrDefaultAsync`/`SingleOrDefaultAsync` for a single conditional result. All are async, so they don't block the calling thread while the database responds.